Anomaly detection using surprisals

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Abstract

TBC

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1 Introduction

Anomaly detection methods are often rather ad hoc and lack a solid theoretical foundation, or they make strong distributional assumptions that may not hold in practice. Here we propose a general framework for anomaly detection using extreme value theory to estimate the probability of each observation. This approach allows for a unified treatment of anomaly detection in both univariate and multivariate settings, and can be applied to both unconditional and conditional distributions. The basic idea is to calculate the surprisal (equal to the negative log density) of each observation, and then to use extreme value theory to model the tail of the surprisal distribution. This allows us to estimate the probability of observing a surprisal value as extreme or more extreme than a given observation, which can be used to flag anomalies. The surprisal approach to anomaly detection provides a principled way to control the false positive rate when flagging anomalies, by setting a threshold on the estimated tail probabilities.

We show that under weak conditions on the tail of the surprisal distribution, the maximum surprisal converges to one of the three types of extreme value distributions. This justifies fitting a Generalized Pareto Distribution (GPD) to the largest surprisal values, and using it to estimate tail probabilities. We illustrate the approach with several examples, including both simulated and real data.

Let $\boldsymbol{y}_1, \dots, \boldsymbol{y}_n$ be iid observations on $\mathbb{R}^d$ from a probability distribution F. The *surprisal* of observation $\boldsymbol{y}_i$ is defined as (Stone 2022)

$$s_i = -\log f(\boldsymbol{y}_i),$$

where f is either the density or probability mass function of F. While these are better known as "log scores" in statistics (Good 1952), in this context we prefer the term "surprisal" (coined by Tribus 1961), as it captures the idea that observations with low density are more "surprising" (and therefore potentially anomalous) than those with high density.

Let $G(s) = \Pr\{S \leq s\}$ be the distribution function of the surprisal values, where $S = -\log f(\boldsymbol{Y})$ and

$\boldsymbol{Y} \sim F$. Then, the probability of observing a value with surprisal at least as large as s is given by $1 - G(s)$, which leads to a natural way to assign an anomaly score to each observation $\boldsymbol{y}_i$:

$$p_i = 1 - G(s_i) = 1 - G(-\log f(\boldsymbol{y}_i)).$$

Hence, one way to detect anomalies is to flag observations with $p_i < \alpha$, giving a false positive rate of α if all observations are from F. This is preferable to directly computing the probabilities from the tail of F, as it allows for anomalies to occur anywhere in the distribution of F, not just in its tails. For example, anomalies may lie between modes in a multimodal distribution, or in otherwise low-density regions of the distribution F. It is also preferable to a distance-based approach to anomaly detection (e.g., Wilkinson 2018), as it naturally handles skewed and heavy-tailed distributions.

The surprisal approach converts the problem of working with a potentially multivariate distribution F to working with a univariate distribution G. We can write

$$G(s) = \Pr\{-\log f(Y) \leq s\} = \Pr\{f(Y) \geq e^{-s}\}.$$

This makes clear that the surprisal probability is related to the highest density regions of the distribution (Hyndman 1996). In fact, every observation with $p_i < \alpha$ lies outside the $100(1-\alpha)\%$ HDR region of F.

We will usually not know F, and so we need to estimate it from data or from theory. Such estimates are particularly fragile in areas of low density, which is precisely where we are looking for anomalies.

To address this, Kandanaarachchi & Hyndman (2022) proposed estimating F by a leave-one-out kernel density estimate, and then fitting a Generalized Pareto Distribution (GPD) to the most extreme surprisal values. The underlying idea is that even if F is not particularly well-estimated, the tail of the surprisal distribution should still be well-approximated by a GPD. Hyndman et al. (2025) provides some theoretical justification for this approach in the context of kernel density estimation.

Here we extend that idea to allow F to be any well-behaved distribution, including conditional distributions. Specifically, we fit a GPD to the largest β of $\{s_i\}_{i=1}^n$, obtaining estimates of the location, scale and shape parameters, (μ, σ, ξ). We calculate the probability of each observation or one more extreme, $p_i = \beta\left[1 - P(s_i \mid \hat{\mu}, \hat{\sigma}, \hat{\xi})\right]$, where P is the distribution function of the GPD. Observations not contained in the largest β of $\{s_i\}_{i=1}^n$ are assigned $p_i = \beta$.

To justify this approach, we seek conditions on the surprisal distribution of $S = -\log f(\boldsymbol{Y})$, where $\boldsymbol{Y} \sim F$, similar to the Fisher-Tippett-Gnedenko (FTG) theorem (Coles 2001).

Outline of paper

2 Extreme value theory for the surprisal distribution

We consider three possible conditions on the surprisal distribution, describing increasingly heavy tails: (1) where S is sub-Gaussian (which occurs when f is bounded above and below); (2) where S

is sub-exponential; and (3) where S has a polynomial moment. We state these as three alternative assumptions.

Assumption 1 (Sub-Gaussian)**.** *The random variable $S = -\log f(Y) \in \mathbb{R}$, and satisfies, for all $\lambda \in \mathbb{R}$, and some $\nu > 0$,*

$$\mathbb{E}\exp\{\lambda(S - \mathbb{E}[S])\} \leq \exp\{\lambda^2\nu^2/2\}.$$

Assumption 2 (Sub-exponential)**.** *The random variable S is sub-exponential with parameters ν and b, i.e.,*

$$\mathbb{E}\exp\{\lambda(S - \mathbb{E}[S])\} \leq \exp\{\lambda^2\nu^2/2\} \quad \textit{for all} \quad |\lambda| < 1/b.$$

Assumption 3 (Polynomial)**.** *The random variable $|S|$ has polynomial moments of order $\alpha \geq 1$; i.e., $\mathbb{E}[|S|^p] \leq C^p$ for some $C > 0$ such that $C^p - 1 > 0$.*

Under each of Assumptions 1–3, we seek a result on the limiting distribution of the maximum of the surprisals $M_n = \max\{s_1, \ldots, s_n\}$. If we can show that this distribution falls within the class of extreme value distributions, then we can fit a Generalized Pareto Distribution to the largest surprisal values (Coles 2001).

2.1 S has sub-Gaussian tails

Let's first assume that the surprisals are well-behaved and have a sub-Gaussian tail. Assumption 1 will be satisfied in cases where the density f is bounded both above and below, such as with mass functions on a finite set. However, it may not be satisfied if the log density has no lower bound.

For example, consider the truncated normal distribution on $[a, b]$, where $Y \sim \mathcal{TN}(a, b, \mu, \sigma^2)$ and $-\infty < a < b < \infty$. Then

$$S = -\log(2\pi\sigma^2)/2 - \frac{1}{2\sigma^2}(Y - \mu)^2 - \log\{\Phi(b) - \Phi(a)\},$$

where Φ is the standard normal distribution function. Since $Y \in [a, b]$, S is bounded for all $\sigma > 0$ and hence S satisfies Assumption 1.

Under Assumption 1, we have the following result on the maximum of the surprisals.

Lemma 1. *Suppose $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n$ are iid from a density f, and let $M_n = \max\{s_1, \ldots, s_n\}$ where $s_i = -\log f(\boldsymbol{y}_i)$. If Assumption 1 is satisfied, then, for $s \in \mathbb{R}_+$,*

$$\Pr\{M_n - \mathbb{E}[S] \geq s\} \leq 1 - \left(1 - e^{-\frac{1}{2}s^2/\nu^2}\right)^n.$$

Proof. For any $\lambda > 0$,

$$\begin{aligned}\Pr\{S \geq s\} = \Pr\left\{e^{\lambda(S - \mathbb{E}[S])} \geq e^{\lambda(s - \mathbb{E}[S])}\right\} &\leq e^{-\lambda(s-\mu)}\mathbb{E}e^{\lambda(S-\mu)} \\ &\leq e^{-\lambda(s-\mu)}e^{\lambda^2\nu^2/2}.\end{aligned}$$

Now, we can maximise the RHS of the above bound in λ, which yields the optimal value $\lambda = (s-\mu)/\{M^2\}$. Plugging this in we have the bound

$$\Pr\{S \geq s\} \leq e^{-\frac{1}{2}(s-\mu)^2/\nu^2}.$$

The stated result now follows by noting that

$$\begin{aligned}\Pr\left(M_n - \mathbb{E}[S] \geq s\right) = 1 - \Pr\{M_n \leq s\} &= 1 - \Pr\{S \leq s\}^n \\ &= 1 - (1 - \Pr\{S \geq s\})^n \\ &\leq 1 - \left(1 - e^{-\frac{1}{2}s^2/\nu^2}\right)^n.\end{aligned}$$

□

Corollary 1. *If Assumption 1 is satisfied, then, for all $s > 0$,*

$$\Pr\left(|M_n - \mathbb{E}[S]| \geq \sqrt{2\nu^2 s} + \sqrt{2\nu^2 \log(2n)}\right) \leq e^{-s}.$$

Proof. This follows directly from the union bound:

$$\begin{aligned}\Pr\left(|M_n| \geq \sqrt{2\nu^2 s} + \sqrt{2\nu^2 \log(n)}\right) &\leq n \Pr\left\{|S| \geq \sqrt{2}\nu^2 s + \sqrt{2\nu^2 \log(n)}\right\} \\ &\leq 2ne^{-[\sqrt{2\nu^2 s} + \sqrt{2\nu^2 \log(n)}]^2/(2\nu^2)} \\ &= 2ne^{-\frac{1}{2\nu^2}(2s\nu^2) - \frac{2\nu^2\log(2n)}{2\nu^2} - \frac{\sqrt{2\nu^2 s}\sqrt{2\nu^2\log(2n)}}{2\nu^2}} \\ &\leq e^{-\sqrt{\log(2n)s} - s} \\ &\leq e^{-s}\end{aligned}$$

The first result follows from the iid assumption on S and the union bound, while the second from the assumption that S is sub-Gaussian. The remaining inequalities follow from reorganising terms. □

Corollary 1 is useful as it clarifies that if the surprisal is sub-Gaussian, then the maximal surprisal will follow a reversed Weibull distribution, and thus is within the class of extreme value distributions.

2.2 S has an exponential moment

In cases where the underlying density has full support on $\mathbb{R}$ and the log density cannot be bounded from below, then the sub-Gaussianity condition in Assumption 1 may be violated, as the following result illustrates.

Lemma 2. *If $Y \sim \mathcal{N}(\mu, \sigma^2)$, then $S = -\log f(Y)$ is sub-exponential. In particular,*

$$\mathbb{E}e^{\lambda(S-1/2)+(\lambda/2)\log(2\pi\sigma^2)} \leq e^{\lambda^2/2+(\lambda/2)\log(2\pi\sigma^2)}, \quad \forall|\lambda| < 1/2.$$

Proof. Firstly, note that

$$S = -\log f(Y) = \frac{1}{2}\log(2\pi\sigma^2) + \frac{(Y-\mu)^2}{2\sigma^2}.$$

and that

$$\mu_S = \mathbb{E}[S] = -\int f(y)\log f(y)\mathrm{d}y = \frac{1}{2}\{1+\log(2\pi\sigma^2)\}.$$

Define

$$X = \frac{1}{2}\frac{(Y-\mu)^2}{\sigma^2} - \frac{1}{2},$$

then $\mathbb{E}[X] = 0$. Now, note that

$$\mathbb{E}e^{\frac{\lambda}{2}(S-1)} = e^{\lambda/2\{\log(2\pi\sigma^2)\}}\frac{1}{\sqrt{2\pi\sigma^2}}\int e^{\frac{\lambda}{2}\left[\frac{(y-\mu)^2}{\sigma^2}-1\right]}e^{-\frac{(y-\mu)^2}{2\sigma^2}}\mathrm{d}y.$$

Consider the change of variables $z = (y-\mu)/\sigma$ so that we have

$$\mathbb{E}e^{\frac{\lambda}{2}(S-1)} = e^{\lambda/2\{\log(2\pi\sigma^2)\}}\frac{1}{\sqrt{2\pi}}\int e^{\frac{\lambda}{2}[z^2-1]}e^{-z^2/2}\mathrm{d}y = e^{\lambda/2\{\log(2\pi\sigma^2)\}}\frac{e^{-2(\lambda/2)}}{\sqrt{1-2(\lambda/2)}}$$

for all $\lambda < 1$. Further, it can be verified that

$$\frac{e^{-\lambda/2}}{\sqrt{1-2(\lambda/2)}} \leq e^{\lambda^2/2}, \quad \forall|\lambda| < 1/2.$$

Hence, when we have $Y \sim \mathcal{N}(\mu, \sigma^2)$, then $S = -\log f(Y)$ is sub-exponential with parameter $\nu = 1$ and $b = 2$. □

Therefore, to be able to apply extreme value theory to the surprisal values, we need a version of Lemma 1 that uses sub-exponentiality.

Lemma 3. *Suppose $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n$ are iid from a density f, and let $M_n = \max\{s_1, \ldots, s_n\}$ where $s_i = -\log f(\boldsymbol{y}_i)$. If Assumption 2 is satisfied, then, for $s \in \mathbb{R}_+$,*

$$\Pr\left(M_n - \mathbb{E}[S] \geq s\right) \leq 1 - \begin{cases} (1-e^{-\frac{1}{2}s^2/\nu^2})^n & \text{if } 0 \leq s \leq \nu^2/b \\ (1-e^{-s/2b})^n & \text{if } s > \nu^2/b \end{cases}.$$

Proof. The result follows a similar argument as for Lemma 1, up until we maximize the tail bound. In particular, for any $\lambda > 0$,

$$\begin{aligned}\Pr\left\{S - \mathbb{E}[S] \geq s\right\} = \Pr\left\{e^{\lambda(S-\mathbb{E}[S])} \geq e^{\lambda s}\right\} &\leq e^{-\lambda s}\mathbb{E}e^{\lambda(S-\mathbb{E}[S])} \\ &\leq e^{-\lambda s}e^{\lambda^2\nu^2/2}.\end{aligned}$$

Now, we must consider a constrained optimization to ensure the right hand side exists. In particular, we know from the proof of Lemma 1 that the unconstrained optimum is $\lambda = s/\nu^2$, which corresponds to the constrained optimum on the set $0 \leq t < \nu^2/b$. However, when $s \geq \nu^2/b$, the maximum is

achieved at $\lambda = 1/b$, and putting this together yields

$$\Pr\{S - \mathbb{E}[S] \geq s\} \leq \begin{cases} e^{-\frac{1}{2}s^2/\nu^2} & \text{if } 0 \leq s \leq \nu^2/b; \\ e^{-s/2b} & \text{if } s > \nu^2/b. \end{cases}$$

The stated result again follows by noting that

$$\begin{aligned} \Pr\{M_n - \mathbb{E}[S] \geq s\} &= 1 - \Pr\{M_n \leq s\} \\ &= 1 - \Pr\{S \leq s\}^n \\ &\leq 1 - \begin{cases} (1 - e^{-\frac{1}{2}s^2/\nu^2})^n & \text{if } 0 \leq s \leq \nu^2/b; \\ (1 - e^{-s/2b})^n & \text{if } s > \nu^2/b. \end{cases} \end{aligned}$$

□

Results similar to Lemma 3 also hold for $\Pr\{M_n - \mathbb{E}[S] \leq s\}$, and for $\Pr\{|M_n - \mathbb{E}[S]| \geq s\}$ with an additional factor of 2, i.e.,

$$\Pr\{|M_n - \mathbb{E}[S]| \geq s\} \leq 1 - \begin{cases} (1 - 2e^{-\frac{1}{2}s^2/\nu^2})^n & \text{if } 0 \leq s \leq \nu^2/b \\ (1 - 2e^{-s/2b})^n & \text{if } s > \nu^2/b \end{cases}.$$

Corollary 2. *Under Assumption 2, for $s > \nu^2/b$, as $n \to \infty$*

$$\Pr\left(|M_n - \mathbb{E}[S]| \geq (2b)s + (2b)\log(2n)\right) \leq 1 - e^{-e^{-s}} + o(1/n).$$

Proof. Apply Assumption 2 to see that

$$\begin{aligned} \Pr\left(|M_n - \mathbb{E}[S]| \geq (2b)s + (2b)\log(2n)\right) &\leq 1 - \left\{1 - 2e^{-[2bs + 2b\log(2n)]/2b}\right\}^n \\ &= 1 - \left(1 - e^{-s}/n\right)^n \\ &= 1 - e^{e^{-s}} - \left[\left(1 - e^{-s}/n\right)^n - e^{e^{-s}}\right]. \end{aligned}$$

The result now follows by noting that, for all $s > 0$,

$$\left[\left(1 - e^{-s}/n\right)^n - e^{e^{-s}}\right] = o(1) \leq 0.$$

□

Corollary 2 is useful as it shows that if the surprisal has a sub-exponential tail, then the resulting maximum surprisal tail probability converges to that of an extreme value distribution. In particular, we have that Corollary 2 implies

$$\sup_{s:s>\nu^2/b} \left|\Pr\left\{|M_n - \mu| \geq \left[2bs + 2b\log(2n)\right]\right\} - \left(1 - e^{e^{-s}}\right)\right| = o(1).$$

To see that this implies that $M_n - \mathbb{E}[S]$, appropriately standardized, converges to an extreme value distribution, we first recall that

$$\Pr\{X_n > x\} - \Pr\{X > x\} = 1 - \Pr\{X_n \leq x\} - [1 - \Pr\{X \leq x\}] = \Pr\{X \leq x\} - \Pr\{X_n \leq x\}.$$

Hence,

$$\sup_{x \in \mathcal{X}} |\Pr\{X_n \leq x\} - \Pr\{X \leq x\}| = \sup_{x \in \mathcal{X}} |\Pr\{X_n > x\} - \Pr\{X > x\}|,$$

so that when X_n and X are both continuous we see that Corollary 2 implies that the CDF of $(M_n - \mathbb{E}[S] - 2b\log(2n))/2b$ converges to that of the Type-1 extreme value distribution (i.e., Gumbel distribution); i.e., using Corollary 2 and the above we have that

$$\sup_{s : s > v^2/b} \left| \Pr\left(|M_n - \mu| \leq [2bs + 2b\log(2n)]\right\} - e^{e^{-s}} \right| = o(1).$$

This is one notion of convergence in distribution.

2.3 S has a polynomial moment

Finally, we consider the case where the surprisal has a polynomial moment (e.g., Student-t random variables), and derive a corresponding result to Lemma 1.

Lemma 4. *Suppose $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n$ are iid from a density f, and let $M_n = \max\{s_1, \ldots, s_n\}$ where $s_i = -\log f(\boldsymbol{y}_i)$. If Assumption 3 is satisfied, then, for $s > 0$ and such that $s > C > 0$,*

$$\Pr\left(|M_n - \mathbb{E}[S]| \geq s\right) \leq 1 - (1 - C^p/s^p)^n.$$

Hence,

$$\sup_{s : s > C} \left| \Pr\left(|M_n - \mathbb{E}[S]| \leq (s \cdot n^{1/p})\right) - e^{-s^{-p}} \right| = o(1).$$

Proof. By Markov's inequality,

$$\Pr\left\{|S - \mathbb{E}[S]| \geq s\right\} \leq \frac{\mathbb{E}|S|^p}{s^p} \leq \frac{C^p}{s^p}.$$

To derive the first stated result, again note that

$$\begin{aligned}
\Pr\left\{|M_n - \mathbb{E}[S]| \geq s\right\} &= 1 - \Pr\left\{|M_n - \mathbb{E}[S]| \leq s\right\} \\
&= 1 - \Pr\left\{|S - \mathbb{E}[S]| \leq s\right\}^n \\
&= 1 - \left(1 - \Pr\left\{|S - \mathbb{E}[S]| \geq s\right\}\right)^n \\
&\leq 1 - (1 - C^p/s^p)^n.
\end{aligned}$$

To derive the second stated result, apply the first result with $t = sn^{1/p}C > 1$, which clearly holds for

some n large enough, to see that

$$\Pr\{|M_n - \mathbb{E}[S]| \geq t\} \leq 1 - \left(1 - \frac{C^p}{(Csn^{1/p})^p}\right)^n = 1 - \left(1 - s^{-p}/n\right)^n.$$

Now, recall that, for $x \geq 0$,

$$\lim_{n\to\infty} (1 - x/n)^n = e^{-x}.$$

Since $x \mapsto e^{-x}$ is bounded for all $x \geq 0$, it follows that

$$\sup_{s:s>C} \left|\Pr\{|M_n - \mathbb{E}[S]| \geq t\} - e^{-s^{-p}}\right| = o(1).$$

□

2.4 Three types theorem

Placing the three lemmas together gives the following "three types" Theorem.

Theorem 1. *Let $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n$ be an iid sequence from F, and let $S = -\log f(\boldsymbol{Y})$, where $Y \sim \boldsymbol{F}$. Further, let $M_n = \max\{s_1, \ldots, s_n\}$, where $s_i = -\log f(y_i)$.*

1. *If the surprisal satisfies Assumption 1, then*

$$\sup_{s:s>0} \left|\Pr\left\{\left|M_n - \mathbb{E}[S]\right| \geq \sqrt{2v^2 s} + \sqrt{2v^2 \log(2n)}\right\} - e^{-s}\right| = o(1).$$

2. *If the surprisal satisfies Assumption 2, then*

$$\sup_{s:s>1/b} \left|\Pr\left\{\left|M_n - \mathbb{E}[S]\right| \geq (2b)s + (2b)\log(2n)\right\} - e^{-e^{-s}}\right| = o(1).$$

3. *If the surprisal satisfies Assumption 3, then*

$$\sup_{s:s>C} \left|\Pr\left\{\left|M_n - \mathbb{E}[S]\right| \geq (Csn^{1/p})\right\} - e^{-s^{-p}}\right| = o(1).$$

Theorem 1 gives a corresponding 'three types' theorem for the behavior of surprisals and has a simple interpretation. If the surprisal has Gaussian like tails, then the maximum surprisal is a reversed Weibull; if it only has an exponential moment, then the maximum surprisal will be Gumbel; if the maximum surprisal only has a polynomial moment, then it is Fréchet. As each of these distributions is a member of the generalized extreme value family, then the upper tail of the distribution can be approximated by a Generalized Pareto Distribution (Coles 2001, Theorem 4.1).

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